

Energy-environment-economy nexus in Pakistan: Lessons from a PAK-TIMES model

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ABSTRACT

Pakistan, alongside other developing countries, is struggling to understand, plan and realize the energy-environment-economic nexus. In this study, we develop a national-scale bottom-up energy optimization model (Pak-TIMES) employing ANSWER-TIMES modeling framework which provides insight into this nexus for Pakistan. The reference energy system of Pak-TIMES model provides energy demand, supply, cost of energy and GHGs emissions. In total, five supply side scenarios enacted in this study are BAU (Business As Usual), RE-30 (30% renewables), RE-40 (40% renewables), Coal-30 (30% Coal) and Coal-40 (40% Coal) for the period 2015–2035. The results suggest that the cumulative investment cost incurred will be 170 billion USD to meet the electricity demand (3091 PJ) by 2035 with 181.5 million tons of GHGs emission in the BAU scenario. The most environmentally ambitious pathway, the RE-40 scenario will require 179.2 billion USD for investment with 96.0 million tons of GHGs emission. The Coal-40 scenario will cause the highest amount of GHGs emissions, i.e., 338.9 million tons with an estimated investment cost of 184.8 billion USD. Based on the results of this study, PAK-TIMES model, it is recommended that government needs to invest in renewables; given that, the environmental cost of non-renewable energy solutions is significantly higher.

1. Introduction

Energy resources management and planning is an important component of economic growth which now closely linked with the sustainable development (Cai et al., 2009, 2011). The subject of energy planning assimilated greater attention after the oil crisis of the 1970s. The planning objectives then mainly focused on the provision of ample energy supplies and resource diversifications. However, the energy planning techniques gradually diverted towards determining cost-effective energy solutions considering economic viability as an important parameter (Hussain et al., 2015). Today, energy consumption is an essential parameter for determining people's lifestyles and quality of life worldwide (Anwar, 2016). However, energy transformation to consumption process is always accompanied with the environmental footprints in the form of GHGs emissions which has become a serious

issue globally (Subramanyam et al., 2017a). In order to meet the growing energy demand, a large quantity of fossil fuels consumption is taking place globally, especially in developing countries. This situation is not only contributing adversely to climate change and global warming on a planetary scale but poses severe local impacts by deteriorating environmental quality (Mondal et al., 2010; Tan et al., 2010; Rong et al., 2015). Thus, energy management has become a complex subject in today's world which require careful consideration pertaining economic and environmental cost-benefit analysis following the concept of 3Es, i.e., energy-environment-economic nexus.

The energy and economic planning alongside environmental considerations have become a key focus of policy makers worldwide. The criteria settings required for policy and decision making in this regards takes into account environmental concerns, energy demand and supply, cost functions and variety of constraints altogether. Most of these

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energy planning approaches are now well worked out by employing computer-based energy modeling tools such as MARKal Allocation (MARKal), ENergy and Power Evaluation Program (ENPEP), Long Range Energy Alternatives Planning System (LEAP) (Hussain et al., 2015), Input-Output models and Systemic Dynamics model (Liu et al., 2011). The developed countries are far ahead in addressing the concept of 3Es and have developed their national-scale energy models which are updated regularly, and are actively used for setting economic, environmental and energy security goals. The economy-energy-environment models are in the evolutionary stage, however, some of 3Es model are quite well developed. Nevertheless, every model has specific focus, theme and research approach. As such, all the 3Es models can be classified into two broad categories, i.e., top-down and bottom-up. The bottom-up models follow engineering approach with more technological detail regarding, energy demand, supply, and energy transformation across the system. These models mainly focus on both upstream and downstream activities in the energy system and carry a purpose of environmental impact assessment (Liu et al., 2011).

Unfortunately, Pakistan is confronting worst energy crisis in the form of electricity and natural gas shortage which has led to the policy of forced load shedding in the country (Aized et al., 2017). The electricity supply outages in the country over the last decade ranged from 8 to 12 h per day in urban areas and up to 18 h in rural areas. The reasons behind the failure of the energy system are the growing gap between demand and supply, followed by depleting energy resources and financial constraints (Valasai et al., 2017). It is, however, important to note that Pakistan so far lacks the appropriate application of energy modeling tools towards integrated energy planning and energy policy formulation. Meanwhile, disintegrated energy planning at various levels has fetched country in a state of energy crisis. There are only a few studies found in the literature in the Pakistan's context which too mainly by academia (Anwar, 2016; Harijan, 2008; Perwez et al., 2015; Rehman et al., 2017a; Valasai, 2017) and (Rehman et al., 2017b) wherein energy modeling tools have been used. There is only one integrated energy planning effort at government level which was in fact in the shape of Technical Assistance (TA) from Asian Development Bank (ADB) in 2007 (IRG, 2010). This study used International Energy Agency's (IEA) TIMES energy model, however, the PAK-IEA model of this study also could not find any connection to the policy formulation. As such, none of the energy policies announced by country so far reflect the finding of energy modeling exercise. It is, therefore, high time that Pakistan employ these modeling tools for a planning paradigm and energy policy analysis that should deliver integrated planning agenda not only to help to overcome electricity crisis (Mirjat et al., 2017) but to guide the country towards cleaner and cheaper energy.

Hence, in this study we developed a national-scale bottom-up

energy optimization model for Pakistan using ANSWER-TIMES (The integrated MARKal EFOM system) modeling framework named as Pak-TIMES model. This attempt is first of its kind that discusses energy, economics and environment nexus in detail in the context of Pakistan. The novelty of this study further lies in the fact that TIMES-model has been used for the first time towards understanding the energy, environment, and economy nexus on a countywide scale in Pakistan. This also demonstrates the capability of the Pak-TIMES model which has been developed in a data-scarce environment and is, therefore, can be refined with the sufficiency of data in future. However, due to the scope of this paper, we have mainly presented results and outcomes of Pak-TIMES which hold importance for policy lessons pertaining energy-environment-economic nexus. Electricity is the major form of final energy used in the country and produced by taking some quantity from all forms of primary energy resources. Therefore, the cost functions and economic details alongside environmental emissions (GHGs) pertaining to electricity demand, supply and generation have been provided in greater details. The forthcoming sections of this paper consist of overview of energy demand and supply in Pakistan (Section 2), power and energy demand modeling in Pakistan (Section 3), energy-environment-economic nexus in Pakistan (Section 4), methodological framework followed in this study (Section 5), results and discussion (Section 6) and finally conclusion and policy recommendations (Section 7).

2. Overview of energy demand and supply in Pakistan

The Hydrocarbon Development Institute of Pakistan (HDIP), the ministry of petroleum and natural resources is the key department responsible for energy data record keeping and management. The figures published by the HDIP reported that the total primary energy supplies back in 2015 was 70 million TOE. In fiscal year 2014–15, oil, natural gas, LPG, LNG (imported), coal and electricity (imported) shared 35.5%, 42.7%, 0.7%, 7.0% and 0.2% respectively. The oil production rate was 94,493 barrels per day in the year 2015, while natural gas produced was 4016 MMCFD in 2014. The same year, 47 exploratory wells were drilled in total along with the drilling of 35 appraisal/development wells while the country paid USD 12 billion for the import of oil. Similarly, in the base year 2015, new hydropower projects of worth 438 MW were added to the national grid which raised the country's total electricity production potential to 106,966 GW h with varying share for different types of the power plants (Thermal: 63.5%, hydel: 30.4%, nuclear: 5.4%, other renewables: 0.7%) (HDIP, 2016). It is significant to note that oil and petroleum products have had been dominating the total energy demand and supplies across the various sectors of the economy since 1992. The demand for oil is mainly coming from the transport sector, while natural gas is a predominant fuel in

Table 1
Energy fuel mix across various sectors from 1992 to 2015.

Fuel Type		Domestic	Commercial	Industrial ^a	Agriculture	Transport	Other Govt.
Oil	1992	8.56%	0.42%	15.48%	3.39%	68.25%	3.90%
	2000	4.12%	0.00%	17.35%	2.55%	72.97%	3.01%
	2015	0.66%	0.00%	9.45%	0.28%	86.84%	2.77%
Natural gas	1992	31.48%	5.81%	62.70%	0.00%	0.01%	0.00%
	2000	40.47%	6.28%	52.55%	0.00%	0.70%	0.00%
	2015	41.30%	5.23%	43.60%	0.00%	9.88%	0.00%
Coal	1992	0.22%	0.00%	99.78%	0.00%	0.00%	0.00%
	2000	0.04%	0.00%	99.96%	0.00%	0.00%	0.00%
	2015	0.00%	0.00%	100.00%	0.00%	0.00%	0.00%
Electricity	1992	33.82%	6.33%	36.27%	17.26%	0.09%	6.23%
	2000	47.06%	5.58%	28.96%	9.96%	0.03%	8.40%
	2015	48.30%	7.59%	29.11%	9.36%	0.00%	5.64%
LPG	1992	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
	2000	75.00%	25.00%	0.00%	0.00%	0.00%	0.00%
	2015	41.33%	41.16%	0.00%	0.00%	0.00%	17.51%

^a Power generation has been included in industrial sector; Other Govt.: Other Government Sectors.

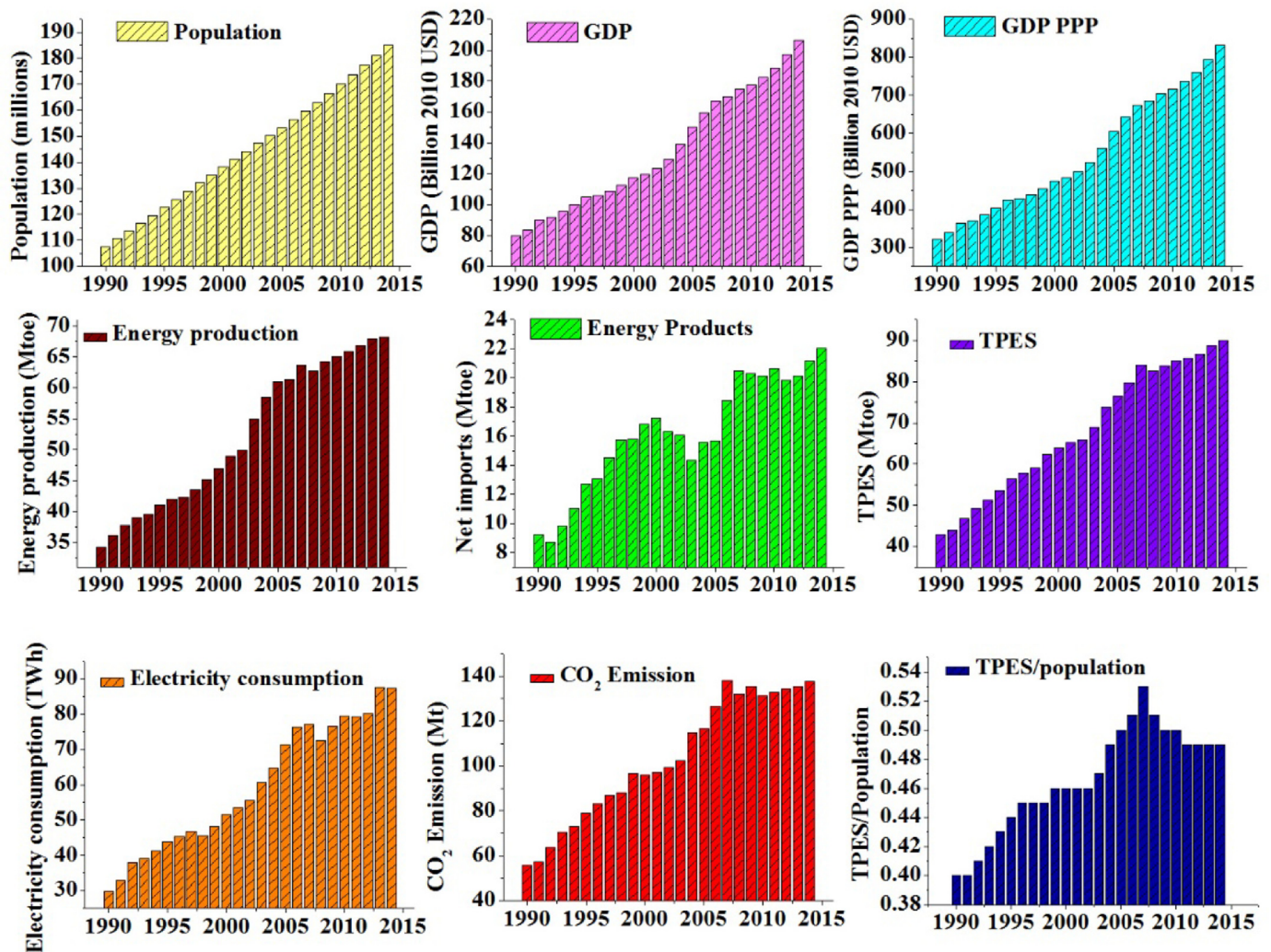


Fig. 1. Energy and socio-economic indicators of Pakistan.

domestic, power generation and industrial sector. Due to changes in government policies and decisions, the variations in energy fuel mix of the country are evident (Table 1). Based on the observed inclination of policy and decision makers towards more oil and hydrocarbons, essential reforms in the power generation mix are inevitable so that the dependency on imported oil and gas can be avoided (Kessides, 2013; Mirza et al., 2007). Owing to these huge imports of fossil fuels, Pakistan's energy security is at a serious risk for the energy price-shocks and varying global energy supplies chains. This dictates that the country needs to revive the indigenous energy production policy, develop efficient energy systems and avoid energy losses to be able to get rid of the import cost for energy commodities (Shahzad and Jamil, 2015).

With the current rate of economic growth, it has been anticipated that Pakistan's energy growth in future will be 8.8% which will raise the country's energy demand to 361 MTOE by 2030. This implies that Pakistan has to raise energy supply up to six times the supplies of 2010–11 (Khan et al., 2016). Pakistan has so far not been able to diversify the energy supply options, mainly belated introduction of renewable energy policy by the government. The maiden renewable energy policy was adopted in 2006; however, the policy has not yielded some promising results so far (Javaid et al., 2011). Moreover, another policy was introduced by the government in 2013 which aimed at addressing the needs for present and future energy demand (Rauf et al., 2015). Unfortunately, despite these efforts, the forced electricity load shedding in 2013 was about 16–18 h per day in various grid-tied areas

of the country (Khokhar et al., 2015).

One of the reasons, the government could not catch up the energy supply targets was the lack of long-term plans based on precise estimates of future energy demand. It is mainly because that there existed so far no solid plans which the authorities in the government could implement to avert the energy crisis. Furthermore, an almost negligible piece of research was undertaken by the government departments on energy planning and policy development. Only, a handful of studies and reports have been documented by the government which merely contains the energy demand-supply forecasts alone. However, these studies are not enough and contain cursory details. For instance, the total energy supply forecasted by GOP (2005) for the year 2015 was 120.8 MTOE, which remained only 70 MTOE although no measure were taken beforehand to ensure essential supplies.

3. Power and energy demand modeling efforts in Pakistan

One of the reasons, Pakistan has not been able to achieve energy security is the lack of serious energy planning efforts based on energy models for sustainable energy resources management and policy development. In the energy planning, the energy models are developed and updated regularly, and they help greatly in the decision making the process at each and every step of energy planning. In the developed world, the central ministries with the help from academia undertake the energy modeling studies based on various methodologies, such as

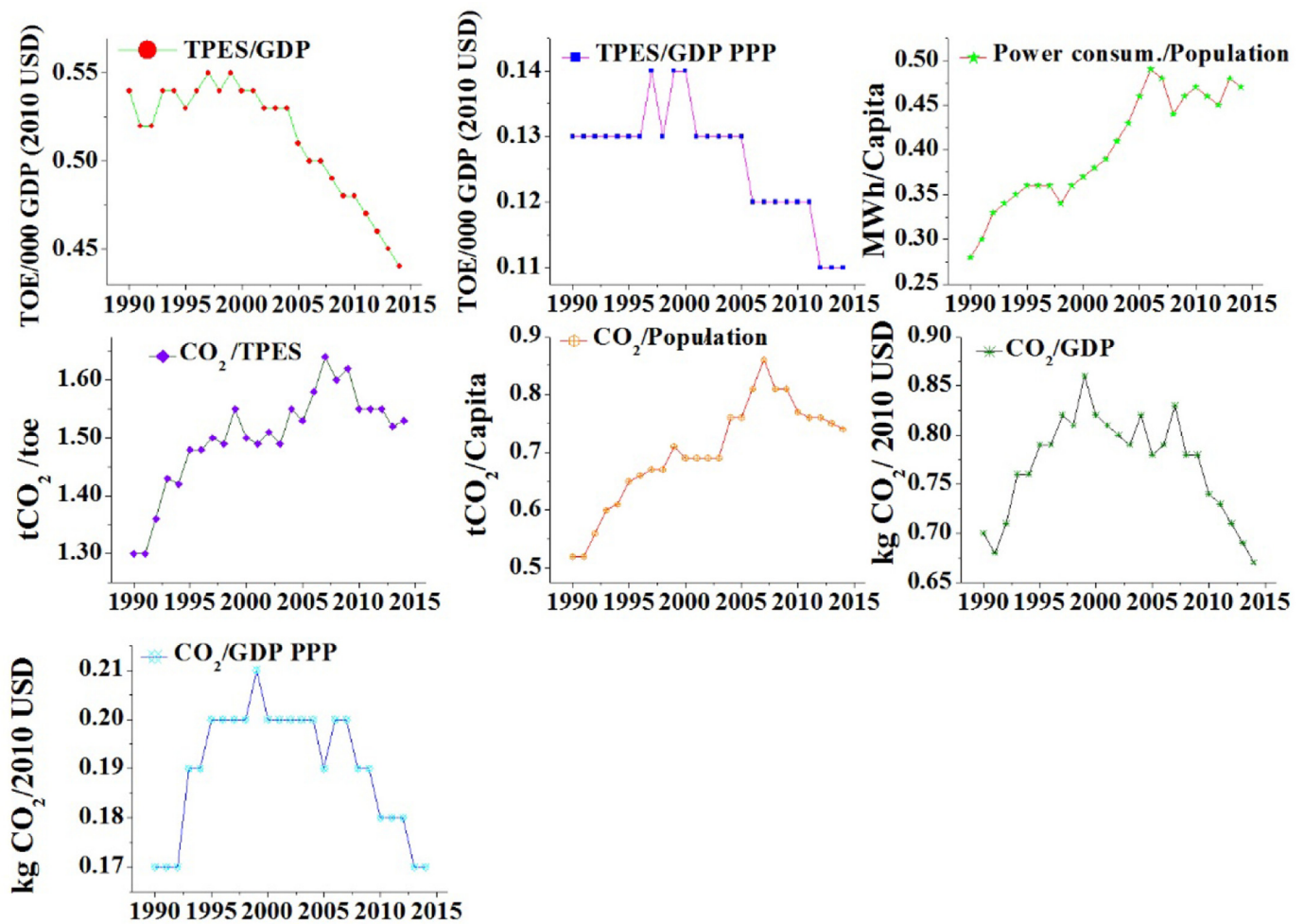


Fig. 2. Energy-environment indicators in Pakistan.

econometrics, simulation, and optimization. Unfortunately, the subject of energy modeling in Pakistan has been given little or no attention. Therefore, there are a handful number of studies in the existing literature that could be referred to such as Rehman et al. (2017a), Rehman et al. (2017b), Hussain et al. (2016), Perwez et al. (2015), Perwez and Sohail (2014), International Resources Group (2010), Anwar (2010), Anwar (2016), GOP (2005), Petroleum Institute of Pakistan (2015), Harijan (2008) and NTDC (2014). Further, a narrowed insight disclosed that most of these studies have focused on electricity demand and supply only; e.g., NTDC (2014), Hussain et al. (2016), Perwez et al. (2015), Valasai (2017) and Perwez and Sohail (2014). Unlikely, as already mentioned above, in the developed countries energy models that are kept live endeavors, updated regularly and are used in the energy planning process which influences the government policy towards the whole economy. In case of Pakistan, we have few studies that somewhat addresses energy flow across all sectors and takes into account multiple fuels such as Rehman et al. (2017a), Rehman et al. (2017b), Anwar (2016), GOP (2005), Petroleum Institute of Pakistan (2015), Anwar (2010) and International Resources Group (2010). However, none of these studies has been pursued by the government towards a commendable outcome and ownership from the authorities concerned. Herein, we provide some details about the existing studies made to forecast power and energy demand in the country.

The government-owned studies include the attempts made by Water and Power Development Authority (WAPDA) which for long administered the power planning and management activities. The power planning wing of WAPDA made the first comprehensive power plan for

the country in 1994. The plan addressed load forecast, generation planning, and transmission expansion planning for the period 1992–2018. This plan was updated further in 1995, 1996, 1998, 2003, 2008, 2011, 2012 and in 2014. The latest regression based load forecast report in this continuation was prepared by NTDC (2014) in 2014 and is based on a complete review of historical data which encompasses electricity consumption, electricity tariff, GDP, and population, etc. The forecast extends to the year 2037 and incorporates three scenarios, i.e., low (3.56% GDP growth), normal (5% GDP growth) and high (6.5% GDP growth). The independent parameters considered in the regression analysis consisted of, sectoral GDP (agriculture, manufacturing, services, trade, etc.), category-wise electricity demand, tariff-wise electricity prices, category-wise customers and load shedding (both PEPCO and KESC), Consumer Price Index (CPI) and population etc. Thus, electricity consumption (GWh) for various consumer categories including domestic, commercial, industrial and agricultural were selected as dependent variables. A total of 51 different regression models were developed and tested with a different combination of response variables. Electricity demand in each sector was forecasted using multiple regression, involving estimation of elasticity coefficients using Ordinary Least Squares (OLS) technique. Petroleum Institute of Pakistan (2015) attempted to develop energy outlook for Pakistan; this study followed regression analysis that consisted of single and multiple response variables. Energy demand forecast was made for industrial, residential, agriculture, commercial, and transportation sectors till 2025. The study followed trial and error approach in order to choose the best fit variables and discard parsimonious and unreasonable variables. In some cases, the last observation's residuals (difference of predicted and actual

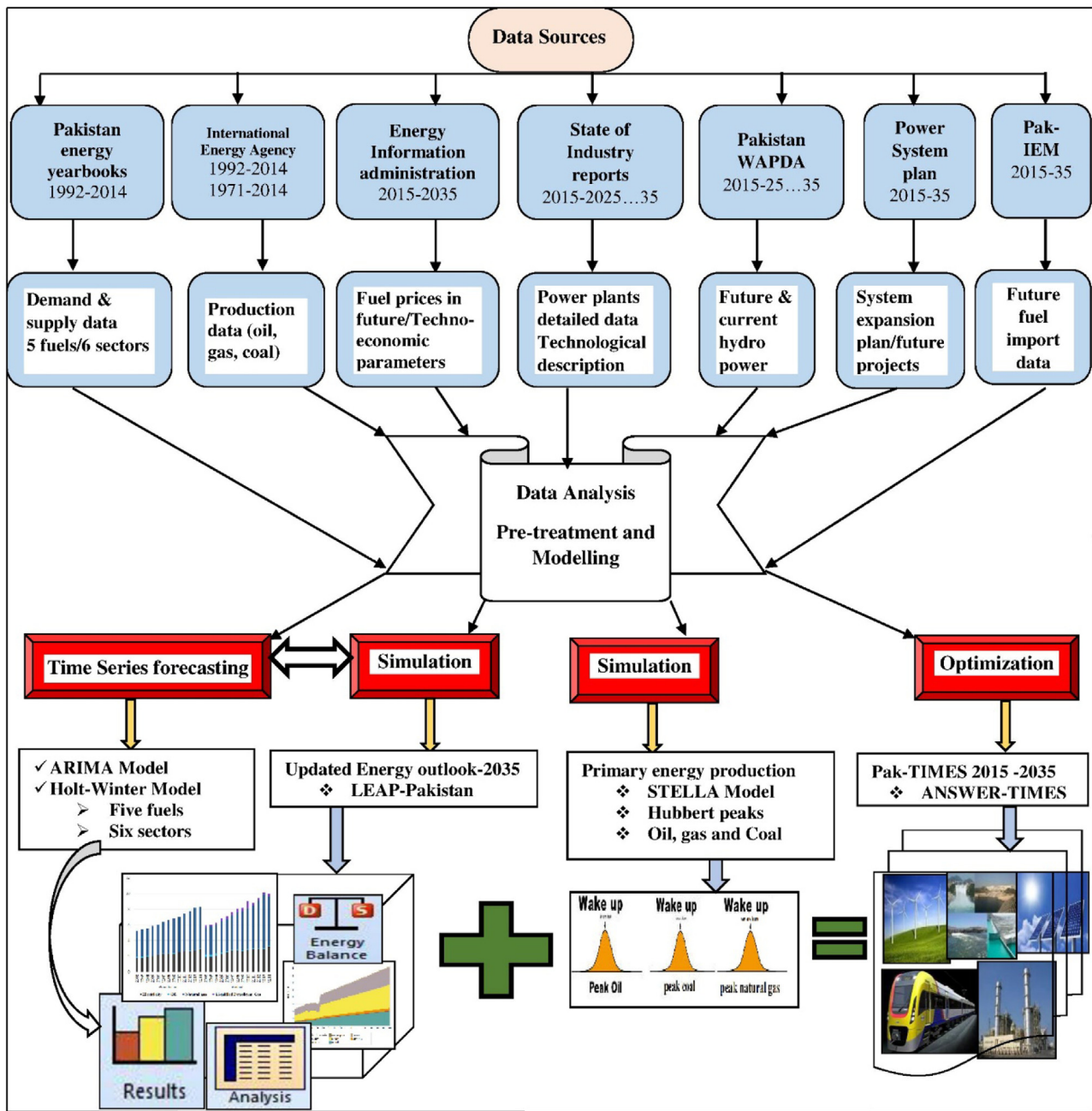


Fig. 3. Methodological flowchart depicting modeling approaches followed.

numbers) were to smoothen the last years' effect.

Amongst academia, [Perwez and Sohail \(2014\)](#) analyzed the demand outcome of various long-term energy pathways by adopting long-range energy alternative planning (LEAP) model to forecast electricity demand and supply for the period 2011–2030. The study further developed three supply-side scenarios to meet the demand which are; i) Business as Usual (BAU), ii) New Coal (NC) and iii) Green Futures (GF) scenarios. Since the final energy intensity data was not available for Pakistan; the study relied on total yearly demand for electricity and number of the consumer in each sector for the estimation of final energy intensity. Similarly, [Anwar \(2010, 2016\)](#); forecasted energy service demand following three different techniques: i) econometric models ii) relating energy service demand in particular sector with respect to GDP and iii) relating energy service demand to value added of the particular sector. For each sector, different approaches were adopted, i.e., for energy demand in transport and residential sector econometric

approach was used by taking a number of energy devices, passenger per km, ton-km as dependent variables; while GDP and population as independent variables. Unlikely, future energy demand in industrial, commercial and agriculture sectors was estimated through GDP and economic value-added approach. Hence, the energy demand in each sector in particular year was estimated by multiplying energy demand of that sector in the base year to the ratio of current and base years GDPs. On the other hand, energy demand service in the agriculture, commercial and industrial sector were estimated by multiplying energy demand of the particular sector in the base year to the ratio of the current and base year value added. [Valasai \(2017\)](#) used dynamic energy economic model for electricity forecasting. While [Hussain et al. \(2016\)](#), adapted ARIMA and Holt-Winter time series forecasting models for power demand forecasting across all sectors of the economy. Both ARIMA and Holt-Winter models along with LEAP were used by [Rehman et al. \(2017a\)](#) for energy demand forecasting across 6 sectors and 5 fuels

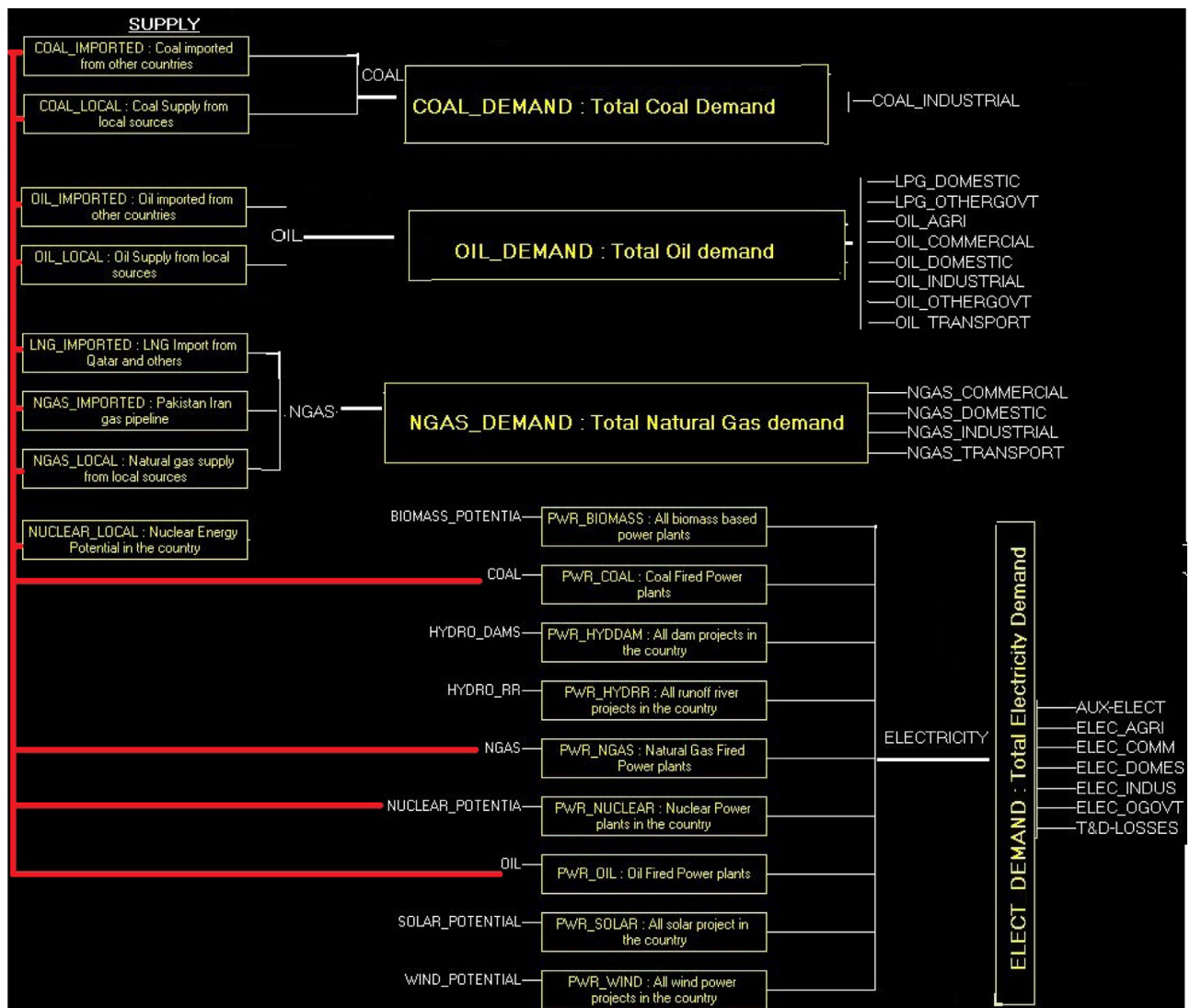


Fig. 4. Reference energy system of the Pak-TIMES model.

Table 2
Electricity production (PJ) and technological mix in BAU Scenario.

Technologies	2015	2020	2025	2030	2035
Biomass-based Power plants	1.15	17.67	31.78	45.89	210.00
Coal fired power plants	0.52	154.76	219.84	284.92	556.06
Hydropower plants (dams)	113.42	174.91	300.00	495.00	680.00
Hydropower (Run-of river plants)	3.48	43.67	124.58	350.00	575.00
Natural gas run power plants	102.06	170.22	180.15	190.07	335.00
Nuclear power plants	20.89	37.93	69.05	96.50	265.00
Oil-run power plants	141.79	106.34	70.90	35.45	–
Solar power plants/stations	0.09	18.63	77.91	159.32	270.00
Wind power plants/stations	5.00	26.00	73.04	155.00	200.00

Table 3
Investment cost for various technologies in BAU scenario (Billion 2015 USD).

Technologies	2015	2020	2025	2030	2035
Biomass-based Power plants	0.20	1.46	2.48	3.58	16.39
Coal fired power plants	0.27	24.29	24.29	24.29	31.66
Hydropower plants (dams)	7.44	10.95	43.47	45.27	45.27
Hydropower (Run-of river plants)	0.15	1.82	4.53	12.74	20.93
Natural gas run power plants	3.93	5.68	5.68	5.68	5.68
Nuclear power plants	2.29	6.84	15.12	20.64	29.02
Oil-run power plants	4.45	4.45	4.45	4.45	4.45
Solar power plants/stations	0.13	3.49	3.49	6.66	11.29
Wind power plants/stations	0.24	2.15	2.15	4.56	5.88

in Pakistan. The future of primary energy supplies (coal, natural gas, and oil) was projected by Rehman et al. (2017b), using STELLA (Systems Thinking, Experimental Learning Laboratory with Animation) model based on total estimated reserves and constant production rate.

The above-discussed studies are pertinent in Pakistan's context; however, these studies yet could not inspire the policy and decision makers in the government for various unclear and unknown reasons.

4. Energy-environment-economic nexus in Pakistan

Economic growth is a function of both renewable and non-renewable energy consumption alongside other production factors, e.g., capital and labour. The mutual correlation between energy and economic growth has been defined by the theorists under four hypotheses that holds for every economy. These hypotheses are neutrality hypothesis, feedback hypothesis, conservation hypothesis and growth hypothesis (Shahbaz et al., 2012). Energy demand in a particular economy is linked to many factors, e.g., technological availability and life quality,

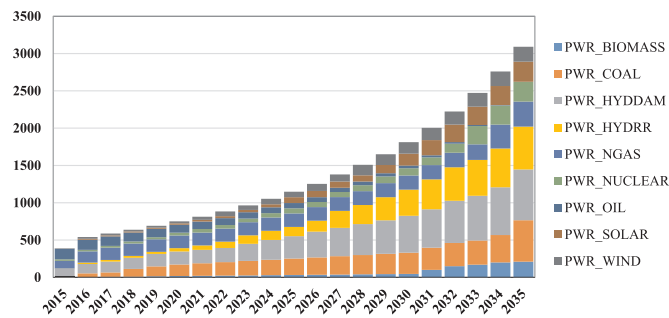


Fig. 5. Increase in electricity generation (PJ) using various technologies (BAU scenario).

Table 4

Total investment cost^a of electricity sector across various scenarios.

Scenarios	2015	2020	2025	2030	2035
BAU	19.11	61.14	105.68	127.88	170.58
Coal-30	19.94	61.14	105.68	125.56	178.73
Coal-40	21.05	61.14	105.68	134.87	184.86
RE-30	19.11	62.39	107.45	132.15	173.33
RE-40	19.32	63.26	111.30	139.67	179.26

^a Billion 2015 USD.

Table 5

GHGs emission across various scenarios in Pak-TIMES (kTonne).

	BAU	Coal-30	Coal-40	RE-30	RE-40
2015	39,439.58	44,256.87	49,207.23	39,439.58	39,439.58
2020	80,052.37	88,405.54	97,966.33	65,327.95	53,347.15
2025	91,022.97	108,093.82	122,716.00	83,415.92	65,087.02
2030	101,993.56	167,944.80	214,138.33	101,993.56	79,869.77
2035	181,518.23	260,128.47	338,922.77	143,200.83	96,014.37

real income, economic structure and relative prices (Khan and Ahmad, 2008). Therefore, economic growth is significantly correlated with energy consumption, i.e., greater is the demand for energy the higher will be economic growth and vice versa (Alam et al., 2007). Pakistan is a developing country with rising energy consumption per capita, i.e., energy intensity is growing with time (Fig. 1). At the same time, the country's dependency on imported oil and petroleum products is also increasing. Thus, Pakistan is facing challenges with respect to fulfilling the growing energy demands due to limited indigenous potential (Aqeel and Butt, 2001).

The economy of Pakistan showed steady GDP growth of 5.4% during 2001–2007 with the highest GDP growth of 8.9% in 2005. However, the GDP growth dramatically declined to a surprisingly low value of 0.36% in 2009; hence, energy consumption dropped down from 39.4 MTOE in 2008 to 37.3 MTOE in 2009. This implies that, energy consumption is a key indicator of a country's stable economy and that both are strongly interlinked functions (Ahmed et al., 2015b), which can be observed throughout the history of Pakistan (Fig. 2). Alahdad (2012) proposed that with an annual economic growth rate of 6.5%, the estimated energy demand would be 198 MTOE in 2025.

There exist numerous studies in the literature wherein the relationship of economic growth with various components of the energy-environment system in Pakistan is discussed in depth. These studies have investigated the correlation between energy demand of individual fuels (oil, natural gas, coal, electricity) and different indicators of the economy such as GDP growth, foreign direct investment, energy prices, etc. Most of the authors have tested the classical hypotheses of energy-economic growth nexus in the country such as Aqeel and Butt (2001), Siddiqui (2004), Nasir and Ur Rehman (2011), Khan and Ahmad (2008), Shahbaz and Lean (2012), Hye and Riaz (2008), Shahbaz et al.

(2012), Zaman et al. (2012), Raza et al. (2015), Ahmed et al. (2015a), Komal and Abbas (2015), Shahbaz et al. (2013), Shahbaz et al. (2015) and Arshad et al. (2016).

Another concern emerging with the economic growth and thus energy consumption is the deterioration of environmental quality. Therefore, several researchers have investigated the emissions of GHGs as an outcome of economic growth and energy consumption. Alam et al. (2007) find out that 1% rise in economic growth leads to 0.84% increase in CO₂ emission in long-run; while 1% increase in CO₂ emission and energy intensity will cause the level of development to rise 1.2% and 0.3% respectively. Ali and Abbas (2013) assessed CO₂ emission from energy consumption both in terms of direct and indirect emissions. Nasir and Ur Rehman (2011) estimated that 1% rise in per capita real GDP would raise per capita emissions by 7.20% in the long-run. Likewise, 1% growth in per capita energy demand leads to 1.65% rise in per capita carbon emissions. This implies that energy consumption in certain sectors might cause an increase in CO₂ emissions but not necessarily economic growth such as in the case of transport and residential sectors. Besides, the relationship between CO₂ emission, economic growth, and energy consumption is positive. However, both residential sector and transport sector includes activities which can be regarded as non-economic activities and hence does not contribute to economic growth. On the other hand, energy consumption in the two sectors combined is almost half of the overall energy demand in the country. Therefore, it is important to improve energy efficiency in these sectors, i.e., strongly encourage solutions which are cost effective and less energy intensive.

5. Methodological and theoretical framework

Energy planning is an integral part of the socio-economic planning and development in a country that guides the development process towards improving life quality. There are various tools used for energy planning worldwide. MARKAL energy modeling and optimization framework is popular energy planning methodology adopted by more than 70 countries and institutes around the world (Al-Riffai et al., 2017; Mondal, 2010; Mondal et al., 2017; Wright et al., 2010). MARKAL is an acronym for “MARKet Allocation” which was developed in 1974 following the oil crisis by International Energy Agency (IEA) based on the General Algebraic Modeling System (GAMS) (Mondal, 2010). The modeling scheme in MARKAL allows the user to connect both supply and demand side of the energy system, enabling the modeler for adding macro-economic data at the same time. Hence, it promises both engineering and economic aspects of the energy accounting and modeling efficiently (Anwar, 2016). Due to detailed parameterization in the MARKAL model, it is classified as a bottom-up energy optimization modeling framework which is capable of accounting environmental impacts in the form GHGs emission (Subramanyam et al., 2017b).

In this study, we have used the ANSWER-TIMES interface of the MARKAL model which is a successor of the MARKAL model with similar principals but with the additional capability for effective use in the Windows-based computer environment. This study model is named as, Pak-TIMES since we have used the TIMES environment of the MARKAL model for Pakistan (Mondal, 2010; Valasai, 2017). TIMES is a linear programming-based optimization model which computes economic equilibrium, i.e., gives optimal results when the end users energy demands are fulfilled employing energy supply sources. During the model running process, TIMES computes solutions where demand is fulfilled at a minimum cost using the available resources (Al-Riffai et al., 2017; Mondal et al., 2011, 2017). However, TIMES is extremely flexible in a way that, it provides the user with the opportunity to set bounds (upper, lower and fix), in order to pick certain resource within limits desired by the modeler. Similarly, the environmental targets based on the choice of technologies can be incorporated by setting out constraints in the model in each scenario as deemed fit (Mondal and Mezher, 2017). The methodological flowchart of the Pak-TIMES model

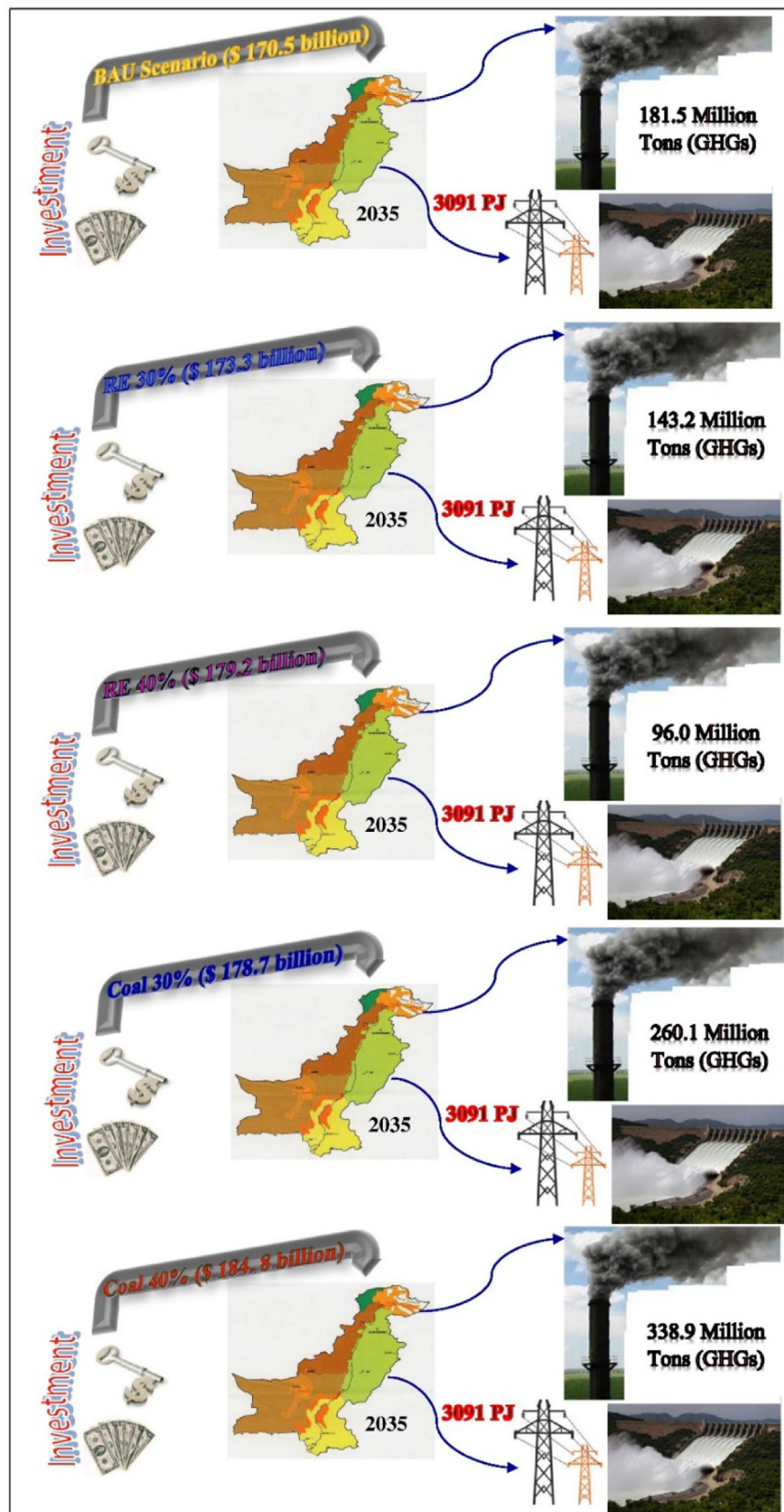


Fig. 6. Investment cost and GHGs emissions across various scenarios.

of this study is shown in (Fig. 3).

In the course of the development of the of Pak-TIMES model, a significant amount of data was used (Fig. 3), besides the data

pretreatment, analysis, and other prerequisites. However, the general procedures and actions followed during formulation of Pak-TIMES model are as under (Mondal et al., 2017; Wright et al., 2010):

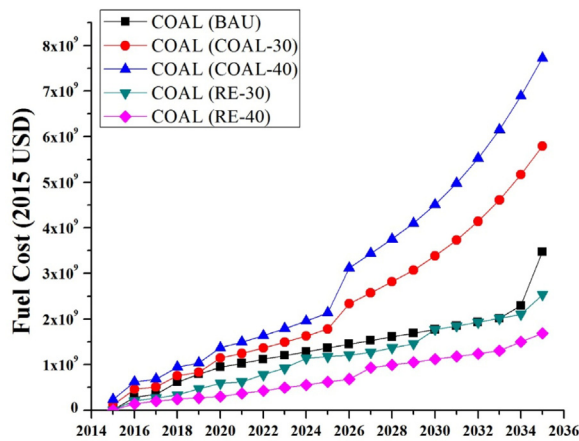


Fig. 7. Input cost of Coal for power generation.

Table 6

Input cost of various fossil fuels for electricity generation (Billion 2015 USD).

Fuels	Scenarios	2015	2020	2025	2030	2035
Coal	BAU	0.00	0.94	1.36	1.77	3.47
	COAL-30	0.12	1.14	1.78	3.38	5.79
	COAL-40	0.23	1.37	2.14	4.51	7.72
	RE-30	0.00	0.59	1.18	1.77	2.53
	RE-40	0.00	0.30	0.62	1.12	1.69
Oil	BAU	3.03	3.20	2.46	1.35	–
	COAL-30	3.03	3.20	2.46	1.35	–
	COAL-40	3.03	3.20	2.46	1.35	–
	RE-30	3.03	3.20	2.46	1.35	–
	RE-40	3.03	3.20	2.46	1.35	–
Natural gas	BAU	1.18	2.61	3.13	3.58	6.76
	COAL-30	1.18	2.61	3.13	3.58	4.03
	COAL-40	1.18	2.61	3.13	3.58	4.03
	RE-30	1.18	2.61	3.13	3.58	6.76
	RE-40	1.18	2.61	3.80	4.31	4.62

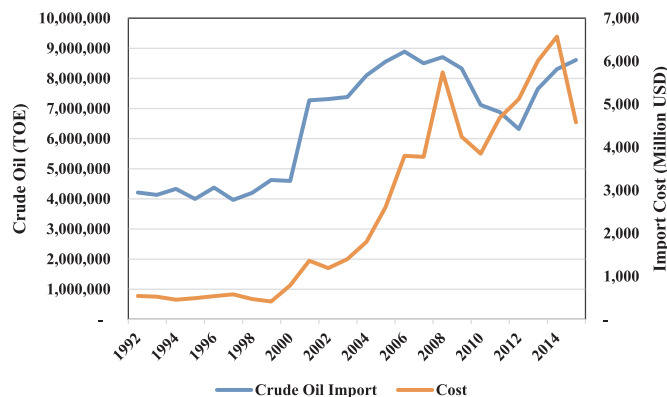


Fig. 8. Crude oil import quantity and revenue spent in Pakistan (1992–2015).

Table 7

Fixed Operation and Maintenance cost (Billion 2015 USD).

	BAU	Coal-30	Coal-40	RE-30	RE-40
2015	0.74	0.76	0.78	0.74	0.75
2020	1.98	1.98	1.98	2.03	2.09
2025	4.29	4.29	4.29	4.37	4.52
2030	5.29	5.08	5.25	5.40	5.67
2035	6.96	7.07	6.83	7.20	7.41

1. Calibration of base year energy balance (consumption and production of all fuels).
2. Accounting primary energy supplies (domestic and import), commodities costs, energy import projects statuses and updates, commencement and design parameters, the status of energy supplies over the planning horizon.
3. Techno-Economic parameters data preparation regarding the current and upcoming power plants, i.e., annual production, installed capacity and capacity addition, availability, efficiency, investment cost, activity cost, operation and maintenance cost (fixed and variable), technical life, pollution and emission coefficients (factors) for each technology, etc.
4. Energy demand projections for all sectors under each category of fuel.

The following section further discusses the development of the reference energy system of Pak-TIMES model.

5.1. The reference energy system of Pak-TIMES model

Energy system mainly comprises of three components which are essentially modelled, i.e. energy intake in the form of primary energy sources, transformation process which converts the primary energy into other forms of energy mainly in final and usable forms of energy, and energy demand devices in the form of sectors and sub-sectors which represents end-users (Fig. 4). This whole setup in a combined-form is called reference energy system (RES) (Mondal and Mezher, 2017). The graphical representation of the RES is a convenient way to understand energy flow and linkages between different components of the energy system (Mondal, 2010; Mondal and Mezher, 2017; Wright et al., 2010).

Based on the specified RES parameters, the model develops equations that consider the whole system altogether. Further, when running the model, the configuration of RES is dynamically adjusted such that, all the objective function value is determined while all other equations (constraints) are satisfied. The usual objective function in case of MARKAL-TIMES is the minimization of long-term discounted cost at a partial equilibrium point between demand and supply (Al-Riffai et al., 2017). Since in both business as usual scenarios (BAU) and other scenarios, the model is required to have anticipated results, therefore, a set of constraints are applied in each case, some of which are general while the others can be specified depending on the objectives of the modeler. Some of these constraints are given as under (Mondal et al., 2011; Valasai, 2017).

- Demand fulfillment at each period in each case;
- Energy balance between demand and supply, i.e., no over or under production;
- Setting limits on environmental emissions when desired, considering emission factors or emission tax;
- The residual capacity transfer for the pre-existing technology and capacity addition in case of new technologies;
- Capacity bounds, activity bounds (energy production) and others, etc.

Following section further, describes the generic details pertaining the Pak-Times model.

5.2. Generic details of the Pak-TIMES model

The Pak-TIMES model of this study has been developed with the spirit to keep this effort live and thus as an updateable model, as such, the technological, spatial and temporal scale of the model can be improved with time. The Pak-TIMES model in the present developed state depicts energy demand-supply across six sectors and five fuels of the economy on a countrywide scale for domestic, industrial, agricultural, commercial, transportation and other government sectors. The fuels

considered are oil, natural gas, coal, LPG, and Electricity. The planning horizon covered by the Pak-TIMES model spans from 2015 to 2035 with 2015 being the base year. The currency used in the model for monetary analysis is the United States dollar based on 2015 exchange rate. The discounting rate applied throughout the planning window is 10% and the energy unit used is peta-joule (PJ). The key energy transformation process considered in the model is the electricity generation as the major form of final energy used in the country. There are nine different electricity generation technologies modelled in Pak-TIMES, which include electricity produced from biomass, solar, wind, hydropower (dams), hydropower (run-of-river), coal-fired power plants, natural gas-run power plants and oil-based thermal power plants. Both energy demand and supply were separately forecasted for the formulation of Pak-TIMES model in initial studies, i.e., in Rehman et al. (2017a, 2017b) using a variety of approaches. However, in the Pak-TIMES model, the electricity demand forecasted by NTDC (2014) has been used as input to the mode. It is because that NTDC forecasts are government estimates and it would be more realistic to address as close as much to the government plans. Similarly, all data sources mentioned in Fig. 3 were followed to gather the required data. The emission coefficient (factors) of about 13 different pollutants in total has been considered including GHGs namely CO₂, N₂O, and CH₄, taken from the online version of GAINS (Greenhouse Gas-Air Pollution Interactions and Synergies) Asia model. The results of Pak-TIMES model pertaining each, business as usual (BAU) scenario, RE-30 (30% renewables without hydroelectricity) scenario, RE-40 (40% renewables without hydroelectricity) scenario, Coal-30 (30% Coal-based electricity) scenario and Coal-40 (40% Coal-based electricity) scenario are presented and discussed in the following section.

6. Results and discussion

The solutions obtained from Pak-TIMES model follows the usual pattern of a standard MARKAL/TIMES model that contains outputs as under (Al-Riffai et al., 2017):

- Investment costs over the planning horizon that denotes the capacity addition (GW) during the specified period;
- Energy fuel mix in each period consisting of optimal contribution from each technology which has been considered,
- Outputs of all transformation processes as PJ, in case of Pak-TIMES generated electricity from each technology;
- Inputs in the form of fuels supplied from both domestic (local production) and outside sources (imports) based on system's requirement at various levels in the RES;
- Outputs in terms of environmental emissions associated with each process provided to the model in the form of emission coefficients;
- The value of the objective function, i.e., the total discounted system cost. In case of Pak-TIMES we have calculated the total system cost of electricity sector by summing up the investment cost, operation and maintenance cost (fixed and variable), fuel cost (both imported and local) and each of which is calculated in the 2015 USD.

The technological choices for electricity generation in the base year 2015 (BAU scenario) consist of oil-run power plants (36.5%), mega hydropower plants (29.2%), natural gas-fired plants (26.8%), nuclear power plants (5.3%), wind power plants (1.29%), micro hydel power stations (0.90), biomass power plants (0.30%), coal-fired power plants (0.13%) and solar power plants (0.02%). However, it is anticipated that the government plans for power generation in future will change significantly which will alter the energy fuel mix in the electricity sector. As such, based on the upcoming electricity generation projects planned by the government, the technological growth rates in the BAU have been updated. It is essentially because the BAU is developed to match the government proposed plans as close as possible. Therefore, at the end of the planning year, i.e., in 2035 the contribution of these

technologies in power generation has changed significantly, and are evident in the results (Table 2). These changes include the increasing dependency on both small scale (run-of-river projects) and large-scale hydropower (dams), the gradual replacement of oil and natural gas-fired power plants by the coal with somewhat equal contribution from wind, solar and biomass energy sources for power generation.

Given above, the aggregate investment cost incurred by the economy will also increase in those cases where the government intends to increase the installed capacity and produce the desired amount of power. We can observe that (Table 3), the investment cost in case of oil-run power plants remains constant over the time. It is because, there is a certain amount of residual installed capacity in the system which once had an investment cost, but, since the capacity is not increasing, so the investment cost observes constant. On the other hand, for technologies like hydropower, coal, solar, wind and biomass the investment cost in future adds up significantly due to gradual addition in the capacity of these plants (Fig. 5).

It is important to highlight that; Pak-TIMES model results pertaining the investment cost across various scenarios is through different but not surprisingly high in scenarios other than BAU scenario (Table 4). The results show that, investment cost increases from 170.58 billion USD in the BAU scenario to 184.86 billion USD in Coal-40, and 179.33 billion USD in the RE-40 scenario during the end year 2035. This relatively small change is owing to the consideration given to the government plans, i.e., capacity addition and change of technologies are not obvious. In the meantime, it is also important to report that documentation regarding future projects are poorly made or entirely not available from the government except for the hydropower project (both small and large scale). Hence, even in the BAU scenario, we have previously directed the model to add a significant quantity of coal, solar, wind and biomass power projects based on the future demand of electricity. As a result, the difference amongst the technological share across various scenarios is not significantly diverse, i.e., the total percentage of renewables (solar, wind and biomass) changes from 15% to 20% in BAU scenario to 20–30% in RE-30 and 30–40% in the RE-40 scenario. Similarly, the compatible investment cost across all scenarios of this study until 2025 is the result of the increase in capacity of coal and renewables made gradually to avoid sudden increase or decrease in technologies which will otherwise be unusual.

Nevertheless, the effect of technological change regarding the environmental impacts is significant (Table 5). It is because, in all scenarios, the electricity production for oil-run power plants has been reduced to zero by the end of the planning horizon, i.e., in the year 2035. The GHGs considered in Pak-TIMES model are CO₂, CH₄ and N₂O and their combined emissions estimated for oil, natural gas, and coal-fired power plants. Since coal is highly polluting fuel with increased CO₂, CH₄ and N₂O emission factors, therefore, increasing the contribution of coal-fired power plants will significantly increase the GHG emissions as evident in both Coal-30 and Coal-40 scenarios (Fig. 6).

The results of Pak-TIMES model immediate seek attention for re-considering the government plans for increasing coal in the energy fuel mix with conventional technologies. From the point of view of investment cost, the RE-40 scenario might not be economical, but, the government plans seem to be in favour of coal because of lower fuel cost (Fig. 7), given that, coal is the most abundantly found fossil fuel in the country.

Over the planning horizon, coal is the only fossil fuel experiencing significant change as a primary energy input for power generation. Therefore, both in Coal-30 and Coal-40 scenario, the coal will cost about 5.79 and 7.72 billion USD respectively in 2035 (Table 6). Conversely, this amount of money might be saved if local coal production starts from Thar coal mines, which are the hugest coal reserves in the country. Crude oil is the most expensive, and imported fossil fuel. As such, if the government stick to its plans of replacing oil-run power plant, 3.03 billion USD can be saved on a yearly basis.

The crude oil reserves in the country are limited, and for a long

time, the government's policy regarding crude oil has been relying on significant quantities of oil import (Fig. 8). Out of the total consumption of oil in the energy fuel mix of the base year 2015, oil-run power plants consumed 38.85% (8.80 MTOE) of oil. The actual import bill for the crude oil was 4.58 billion USD against 8.61 MTOE that accounted for 38.02% of total crude oil consumed in the country (HDIP, 2016). The cost of oil reported in Pak-TIMES, i.e., 3.03 billion USD, is for the oil consumed in power plants only, following, EIA (Energy Information Administration) fuel price projections based on Brent Spot prices. Thus, gradual elimination of oil-run power plants is a good policy if delivered on the ground, given that, the oil is imported from outside and carries a huge cost. But then again, the decision regarding candidature-ship of a substitute-fuel for oil must be made on all grounds of 3Es nexus, i.e., it should be efficient, environmentally clean and economically cost-effective.

The policy of adopting significant coal in the fuel mix may not be sound on economic grounds too, because of the reason that; the operation and maintenance (O and M) cost of coal is similar to the O and M cost of renewables such as for combined wind, solar and biomass plants. Thus, even, if the government attempts to avoid the fuel cost by investing more on coal-based power plants, in the long run, almost the same amount of money shall be spent as operation and maintenance cost (fixed), i.e., 6.83–7.07 billion USD in both coal-based scenarios (Table 7).

Thus, on both environmental and economic grounds the policy of coal-based energy future in the country will not yield promising results. In fact, the air pollution in Punjab province has already been exacerbated, and winter smog in Lahore-Gujranwala-Faisalabad triangle has been termed as the 5th season. In November 2017 alone 18 people have been reportedly killed in smog-related incidents and caused 30 international flights to be canceled at Lahore Airport.

7. Conclusions and policy recommendations

In this study, using ANSWER-TIMES modeling platform a national scale PAK-TIMES model has been developed to analyze the significant relationship and nexus of 3Es (energy, environment, and economy) for Pakistan. PAK-TIMES model results provide very important insights and future policy perspective in the energy management and planning fronts for Pakistan. The study is also significant for the reason that ongoing energy crisis of Pakistan had triggered from electricity shortfall; therefore, energy-environment-economic nexus modeling undertaken is based on the assumption that the electricity generation depicts a major portion of the energy transformation process taking place in the country. Thus, we have provided the total system costs under various fractions such as investment cost, operation, and maintenance cost (fixed and variable), fuel cost, import cost under five supply side scenarios of electricity generation in future alongside the associated impacts in the form of GHGs emissions in each scenario. This study is anticipated to increase the understanding and help the process of decision and policies making regarding energy-environment-economic nexus in the country. The results suggest that the country's future pertaining to environmental quality improvement is highly linked to the decision regarding future energy fuel mix. Pakistan can follow a variety of policies to produce 3091 PJ of electricity by 2035 to meet the demand. However, each policy will cost differently, and the environmental implications will vary accordingly. For instance, in order to catch up the electricity demand, the government is required to invest about 170 billion USD under the BAU scenario that will result into the addition of 181.5 million tonnes of GHGs to the atmosphere. On the other hand, the government can sequester about half of these GHGs, i.e., 85.5 million tonnes by adapting the RE-40 policy determined in Pak-TIMES. Consequently, to do so, the aggregated future investment cost in renewables will be 179.2 billion USD which will equally fulfill the electricity demand. However, if the government employs more coal in the electricity sector with the aim to save the fuel cost, the

environmental quality will deteriorate further. It is due to the fact that, in the Coal-40 scenario, 338.9 million tonnes of GHGs will be released into the atmosphere from the power generation which requires an estimated investment cost of 184.8 billion USD. Hence, this research concludes that the future of energy-environment-economic nexus is highly crucial regarding taking cost-effective and environment-friendly decisions. Research and policy tools, such as Pak-TIMES model carry significance in terms of future energy policy making and planning in the country. The possible policy lessons and recommendations which can be learn from Pak-TIMES model of this study are summarized as under:

- No policy regarding the future energy demand, supply and efficiency improvement should be devised without undertaking energy modeling exercise and decision support systems;
- Energy import projects with the neighboring countries should also be analyzed based on energy modeling thereby considering local energy potential and decreased dependency on outside sources;
- Energy data record keeping, bookkeeping, and sub-sectoral level energy demand-supply should be made available for efficient energy modeling and forecasting studies;
- Access should be provided to the upcoming energy projects with precise techno-economic parameters and detailed description;
- A centralized energy ministry/energy planning wing should be established solely responsible for energy planning and management in the country;
- Manufacturing and inventions regarding energy technologies should be encouraged locally, and funds should be allocated to local developers;
- Energy efficiency improvement programmes should be launched in each sector of the economy thereby discarding all outdated energy demand devices in the residential, industrial, transportation, agricultural and commercial sectors.

It is anticipated that taking in account the findings and recommendations of this study and undertaking further detailed investigation shall improve the on-ground situation and add to maintain an optimal relationship of energy-environment-economic nexus. Balancing this relationship will help in addressing various loopholes in the system, thus guiding the economy towards sustainable energy supplies and thus sustainable development.

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